The rate law can be written as,

$$\text{Rate} = k\,[A]^0$$

$$\frac{-d[A]}{dt} = k\,(1) \qquad (\because [A]^0 = 1)$$

$$\Rightarrow \quad -d[A] = k\,dt$$

Integrate the above equation between the limits of $[A_0]$ at zero time and $[A]$ at some later time 't',

$$-\int_{[A_0]}^{[A]} d[A] = k\int_0^t dt$$

$$-([A])_{[A_0]}^{[A]} = k\,(t)_0^t$$

$$[A_0]-[A] = kt$$

$$k = \frac{[A_0]-[A]}{t}$$

Equation (2) is in the form of a straight line $y = mx + c$

Ie., $[A] = -kt + [A_0]$

$\Rightarrow y = c + mx$

A plot of $[A]$ Vs time gives a straight line with a slope of $-k$ and y - intercept of $[A_0]$.

Examples for a zero order reaction:

1. Photochemical reaction between H_2 and I_2

   $$H_2(g)+Cl_2(g) \xrightarrow{h\nu} 2HCl(g)$$

2. Decomposition of N_2O on hot platinum surface

   $$N_2O(g) \rightleftharpoons N_2(g) + \frac{1}{2}O_2(g)$$

3. Iodination of acetone in acid medium is zero order with respect to iodine.

   $$CH_3COCH_3 + I_2 \xrightarrow{H^+} ICH_2COCH_3 + HI$$

   $$\text{Rate} = k\,[CH_3COCH_3]\,[H^+]$$

General rate equation for a nth order reaction involving one reactant [A].

$$A \longrightarrow \text{product}$$

$$\text{Rate law } \frac{-d[A]}{dt} = k[A]^n$$

Consider the case in which $n \neq 1$, integration of above equation between $[A_0]$ and $[A]$ at time $t = 0$ and $t = t$ respectively gives $\frac{1}{[A]^{n-1}} - \frac{1}{[A_0]^{n-1}} = (n-1)kt$

7.6 Half life period of a reaction:

The half life of a reaction is defined as the time required for the reactant concentration to reach one half its initial value. For a first order reaction, the half life is a constant i.e., it does not depend on the initial concentration.

The rate constant for a first order reaction is given by

$$k = \frac{2.303}{t}\log\frac{[A_0]}{[A]}$$

at $t = t_{1/2}$; $[A] = [A_0]/2$

$$k = \frac{2.303}{t_{1/2}}\log\frac{[A_0]}{[A_0]/2}$$

$$k = \frac{2.303}{t_{1/2}}\log 2$$

$$k = \frac{2.303 \times 0.3010}{t_{1/2}} = \frac{0.6932}{t_{1/2}}$$

$$t_{1/2} = \frac{0.6932}{k}$$

XII U7 kinetics - Jerald Folder.indd 215 3/29/2019 6:35:08 PM